Final Project Design Phase

Computer Science - Week 11

Oct 3, 2026 - Version 0.0.4

For complete instructions and project ideas, refer to the assignment in Canvas. Specifically, review the final project requirements before you continue below. If you have not reviewed the project requirements in Canvas (you might have to go to the modules view and check the current or past week's content), you will not be able to properly complete the parts below. In addition, it may be helpful to review the information on this Drafter documentation page.

0) Project Planning and Preparation

Successful programmers spend time planning before they begin coding. *After* reviewing the project requirements, answer the following questions to help guide your design and development process.

A. What concepts from class or previous assignments might help you complete this project? Which programming concepts (loops, conditionals, lists, classes, etc.) do you expect to use? What programming skills do you want to improve while completing this project?

Probably loops, conditional statements, definitely lists and classes, and pulling data, so routes and functions

B. Which parts of the project do you expect to be the easiest for you? The hardest? Why?

Coding it will probably be the easiest part, but the hardest part is gonna be figuring out what to code because my topic is very complicated

C. Even though you have not yet designed the final project, estimate which tasks will likely require the most time and effort? Which features or components should be implemented first?

The figuring out portion is going to take the most time, but the coding aspect shouldn't be that bad when i figure out what to do

D. What distractions might interfere with your work and how will you minimize them? What resources will you need available while working? What questions, if any, do you still have about the assignment requirements?

I will need to be able to access the stock market, tas, friends for help and definetely an llm

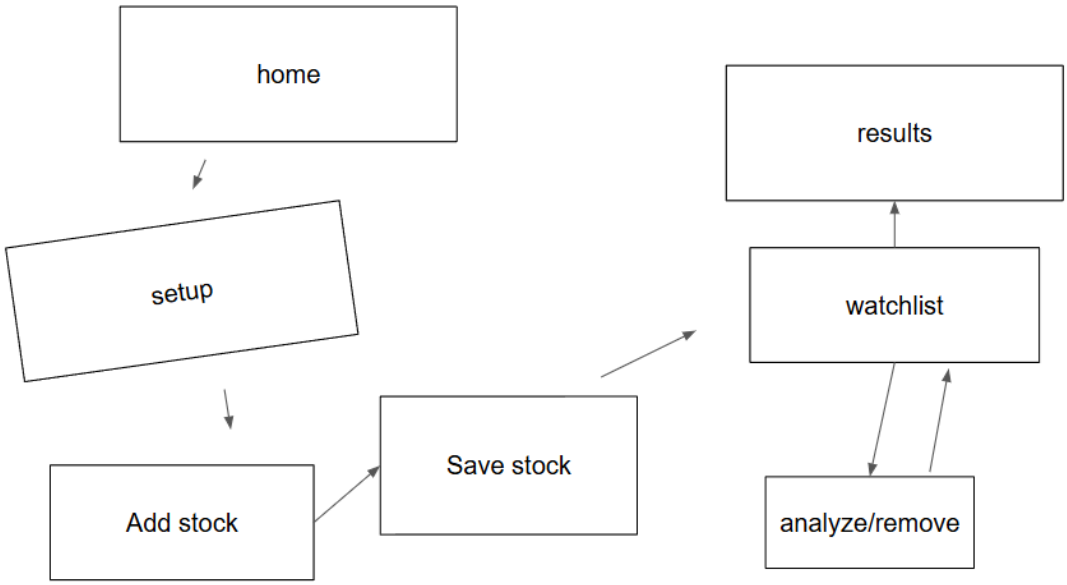
1) Site Diagram

Create a mock-up of your planned website. We suggest you make the mockup on paper, although you can also use software like PowerPoint or Slides to make a mockup. There is other design software that could also be used, but we strongly encourage you to keep it simple. Do not spend more than 15-30 minutes detailing the mock-up. Your goal should be to make a very quick and dirty diagram, to show that you have some kind of idea.

We recommend drawing each page as a rectangle, and laying out the text, buttons, and other components inside. Then, use arrows to indicate links between pages. Label each page the way you will name each function, and then put labels on the arrows to indicate what the button links. Also write down any parameters that should come into the pages.

In addition to the pages and routes, write down what fields your `State` will have, both their names and types. Think clearly about what data will need to be persistent across the entire website.

Add photo(s) or screenshot(s) of your website mock-up and labeled diagram on the next page.



2) Implementing Requirements

This table should serve as a draft for how you will implement the requirements into your final project. You may change your ideas later on, but your final site must fulfill all the requirements.

Requirement	**Implementation**
Purpose/Functionality A 1-2 sentence description of the purpose or functionality of your site	**To help people figure out whether they should invest in a stock or not.**
State Dataclass Attributes Minimum of 4 attributes, one of which must be a list/dict of instances of your second dataclass.	Budget: float The amount someone would want to invest Risk_level: str “Low”, “medium”, or “high” Stocks: list[Stock] Just a list of stocks that the user is interested in

Briefly describe each attribute, its type, and how it will be meaningfully modified throughout the site.	Message: str Feedback or error text
Second Dataclass Your second dataclass, including its attributes, their types, and how they will be meaningfully modified (if they are modified)	@dataclass Class Stock: Name: Price: Low_52wk: High_52wk: Market_cap: Long_term: Verdict: Verdict can change after analyzing
Routes (minimum 7) The names of your route functions	● Index ● Setup_page ● Save_setup ● Add_stock_page ● Save_stock ● Watchlist_page ● Analyze_stock ● Remove_stock ● Results_page
Pages (minimum 5) A brief description of the pages in your site	Home Setup Add Stock Watchlist Results
User Input (minimum 3) Description of user input components (TextBox, TextArea, CheckBox, or SelectBox) and their purposes	Textbox: budget, data/details of stocks Selectbox: Risk level Checkbox: How long do you want to hold it for

For more information on these input types, see Drafter Documentation.	
If Statements (minimum 3) Brief description of what will be evaluated in your if statements and how they will meaningfully change the logic of the site	Affordability: if price > budget, cant afford Value check: if price is in bottom third, invest Risk vs size: if risk level is low and market cap is under a certain threshold, too risky for you
Loop (minimum 1) This loop must iterate over the list/dict of dataclass instances in your State attribute. Description of this loop and why it is meaningful to the site	For stock in state.stocks which builds a table of data/info of a stock

LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

Replace this text with your answer.

Final Submission

When you are finished, download this file as a Word Document (docx) and upload the file to the appropriate assignment on Canvas.